

TechPup — Core Game Loop: Reverse Tamagotchi Blueprint

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Builds on: competitive-analysis-and-usp.md

1. Reverse Tamagotchi Product Blueprint

Core Concept

A classic Tamagotchi asks the owner to feed inputs to a fake pet. TechPup reverses this: the dog's **real collar data** (activity, heart rate, weight, sleep, bark events) determines the avatar's mood. The avatar's needs then become **missions the owner completes with the real dog**. The avatar doesn't need imaginary food — it needs the dog to actually go for that walk.

Why this works where competitors fail

Competitor failure	TechPup answer
FitBark: dashboard with no story, no action	Avatar mood gives the data a face and a daily call to action
Woofz: guilt-based billing, cancellation friction	Shame-free missions — even bad moods reward the owner for showing up
Zigzag: stage-locked, expires after puppy phase	Loop grows with the dog's entire lifespan, all breeds, all ages
Tamagotchi: harsh penalty, no real-world utility	Avatar never "dies" — distressed mood = vet nudge, not game over
Pokémon Go: loop decays without fresh content	Multiple reward horizons + social + seasonal quests sustain long-term return

Design rules (non-negotiable)

- The avatar never dies, never punishes.** Bad data → concerned/distressed mood + vet nudge. Never a game over.
- One primary mission per session.** Mirrors Duolingo's "one lesson." Owner always knows the single next action.
- Rewards for showing up, not for perfection.** A tired or bored mood still earns coins for completing its mission.
- All data reflects the real dog.** Avatar mood is computed from wearable readings — never synthetic timers.
- Cosmetics are earned, never locked behind levels.** Any outfit purchasable from day one with PupCoins.

Core building blocks

Element	Description	Status
Avatar mood	10 mood states derived from <code>compute_wellness_score</code> + breed-specific thresholds	Live
Daily mission	Breed-specific quest per mood (walk, brain drain, recovery setup)	Live
PupCoins	Immediate reward per completed mission	Live
XP + Level	Long-term progression, 50 XP per level	Live
Streak	Consecutive mission days — resets to 1, not 0, on a miss	Live
Badges	Milestone unlocks (3/7/30-day streak, thriving, joint health guardian)	Live
Cosmetic shop	PupCoin-based, no level gates, rotating weekly items	Planned
Passive coins	Every 2 hrs based on avatar mood (happy = coins, thriving = 2×)	Planned
Twice-daily missions	Morning window (06–12) + evening window (17–22)	Planned
Login streak bonuses	Day 7 = +100 coins, Day 30 = +500 coins + exclusive badge	Planned
Dog social meetings	In-person meeting rated by both owners → coins + Social Skills XP	Planned
Training program XP	Completing a lesson awards Training XP + "Good Student" badge	Planned
Vet report	PDF/link export of 30-day history for vet appointments	Planned

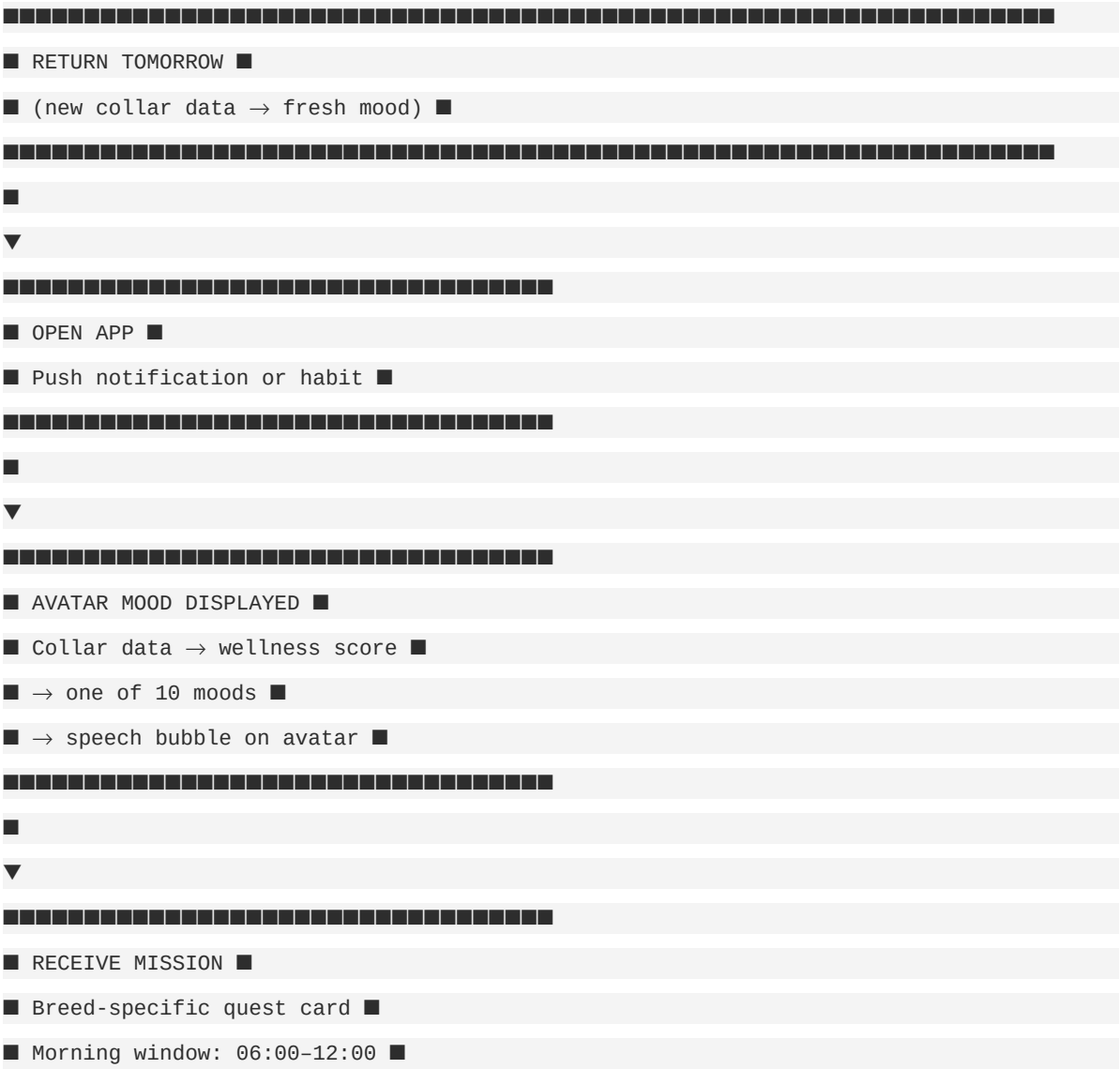
2. User Journey Map

Stage	Trigger	User action	App response	Emotional beat
1. Morning sync	Collar syncs overnight sleep + mobility data	Owner wakes up	Wellness score computed silently in background	—
2. Open App	Push notification: <i>"Tong Gau's avatar wants to see you"</i>	Taps notification	Home tab loads: animated avatar + mood + speech bubble	Curiosity — "how did my dog sleep?"
3. Read the mood	Avatar visible	Reads speech bubble e.g. <i>"I'm bored... let's do 15 min outside!"</i>	Mood explanation shown, quick stats chips loaded	Clarity — "I know exactly what to do today"
4. Accept mission	Mission card displayed	Taps "Start Walk" or reads mission	Mission card shows breed-specific instructions + reward preview	Motivation — "I can do this now"
5. Complete activity with dog	Owner leaves home with dog	Completes walk / brain drain / recovery as instructed	Collar tracks movement; GPS logs route in Activity tab	Bonding — real time with the dog
6. Log completion	Returns home	Taps "Complete mission" in morning or evening window	PupCoins + XP awarded, streak incremented, badges checked	Gratification — coin animation, balloon if badge unlocked
7. Evening check-in	17:00 window opens	Opens app again, logs evening mission if applicable	+Bonus coins if both windows completed today ("Full day")	Consistency reward
8. Passive coins tick	2 hrs elapsed since last open	—	App calculates passive coins based on avatar mood and credits on next open	Surprise — small windfall when opening
9. Level up	Cumulative XP crosses threshold	—	Avatar level-up animation + new cosmetic/quest unlocked	Pride — "my dog's avatar is growing"
10. Browse shop	Earned PupCoins	Opens More tab → Cosmetic Shop	Sees new rotating weekly items; buys an outfit	Personalisation — avatar looks like their dog

Stage	Trigger	User action	App response	Emotional beat
11. Meet a friend's dog	Real-world dog park encounter	Both owners scan QR codes, rate interaction (■/■/♥■)	Coins + Social Skills XP awarded to both; ♥■ triggers fireworks	Social joy
12. Return tomorrow	Evening notification	Loop restarts at Stage 1	Fresh collar data → new mood computed	Habit — automatic, like checking messages

Key insight: Stages 1–2 are passive (the collar and server do the work). Stages 3–6 are the active daily loop the owner controls. Stages 7–12 are the retention layer that gives long-term meaning to the daily repetition.

3. Core Game Loop Diagram





■■■■ RETURN TOMORROW

What makes this "reverse": The loop's energy source is real dog behaviour captured by the collar — not a tap on a virtual food button. The avatar's next mood is only as good as what actually happened with the dog today.

4. Reward System Proposal

4.1 PupCoins — Immediate Reward

Earning sources:

Source	Coins	Notes
Morning mission complete (06–12)	2–20 (mood-based)	See table below
Evening mission complete (17–22)	2–20 (mood-based)	Second window per day
Both windows completed same day	+5 bonus	"Full day" consistency bonus
Passive tick — avatar is happy (per 2 hrs)	3–5	Calculated on next app open
Passive tick — avatar is thriving (per 2 hrs)	6–10	2× multiplier
Day 7 login streak	+100	Large milestone haul
Day 30 login streak	+500	Major haul + exclusive badge + cosmetic
Dog social — Friends (■)	+50	Both owners receive
Dog social — Love (♥■)	+250	Both owners + fireworks animation
Training lesson completed	+10	Per lesson, any program

Mission coins by mood:

Mood	Coins per completion
Thriving	20
Happy	10
Content	8

Mood	Coins per completion
Bored / Tired / Anxious	5
Uneasy / Overtired	2–3
Concerned / Distressed	0

Spending: Cosmetic shop only. No gameplay advantages purchasable. Shop rotates weekly so there is always something new to save toward.

4.2 XP & Levels — Long-Term Progression

- XP per mission completion = `overall_score ÷ 10` (e.g. score 85 → 8.5 XP)
- XP also from: training lessons (+15 per lesson), social interactions (+5/+20)
- 50 XP per level (flat curve for now, adjustable)
- **Leveling never regresses** — a bad week slows growth, never demotes the avatar
- Levels do not gate cosmetics — they gate new quest variants and bonus event access

4.3 Streaks — Habit Reinforcement

Two separate streak counters:

Streak type	What resets it	Milestone bonuses
Mission streak	No mission completed all day	Badges at 3, 7, 30 days
Login streak	App not opened all day	+100 coins day 7, +500 coins day 30

- Missing a mission streak resets to 1 (not 0) — the next action still counts
- Login streak is independent — owner can keep login streak alive even on rest days

4.4 Badges — Milestone Recognition

Badge	Trigger
■ First Steps	Complete first mission
■ 3-Day Streak	Mission streak ≥ 3
■ 7-Day Streak	Mission streak ≥ 7
■ 30-Day Streak	Mission streak ≥ 30
■ Thriving	First thriving mood reading
■ Joint Health Guardian	Recover from concerned/distressed to a good mood

Badge	Trigger
■ Good Student	Complete a full training program
♥■ Love at First Sight	First Love dog social interaction
■ Day 7	7-day login streak
■ Day 30	30-day login streak + exclusive cosmetic

4.5 Unlockable Content — Prevents Loop Decay

The main reason Pokémon Go's revenue halved after 2020 is content stagnation — the loop stayed the same. TechPup counters this with three content channels:

- **New quest variants per level:** e.g. Level 5 Shiba Inu unlocks "advanced scent-trail" brain-drain quests. Keeps missions from feeling repetitive.
- **Seasonal events:** time-limited missions (e.g. "HK Lunar New Year Walk Week", "Summer Hydration Challenge") that replace the default quest for a week. Creates FOMO urgency on a predictable calendar.
- **Breed expansion:** each new breed added to the app is a new content unlock for owners of that breed — new quest copy, new avatar illustrations, new breed-specific badges.

4.6 Dog Social — Interaction Reward

Real-world dog meetings become in-app events rated by both owners:

Outcome	Icon	Coins	XP	Extra
Growled / went separate ways	■	0	+1 Social Skills XP	Social history still recorded
Friends	■	+50	+5	Added to friend list
Love	♥■	+250	+20	Fireworks animation on both screens

- Both owners must confirm — prevents solo farming
- Friend list shows friend dogs' current avatar mood (not raw health data)
- Friends can send one "Paw" reaction per day (+2 coins to recipient)

4.7 Vet-Flagged Moods — The Reward Floor

concerned and distressed moods pay 0 PupCoins and pause passive coin ticks. This is intentional and must not be changed. When a real health signal fires, the game steps aside — a vet nudge must never compete with a reward animation for the owner's attention.

5. Success Outcome

The framework that brings users back daily:

Reward horizon	Mechanic	Frequency
Immediate	PupCoins per mission, passive ticks	Every session (up to 2×/day)
Short-term	Streak counter, badge unlocks	Daily / every few days
Mid-term	XP levels, new quest variants	Weekly
Long-term	Login streak milestones, seasonal events, breed expansion	Monthly

Every morning, fresh collar data produces a new avatar mood. That mood becomes one clear mission. Completing it with the real dog feeds back into tomorrow's data — closing the loop. PupCoins reward the action immediately. XP and levels reward consistency over weeks. Streaks, login bonuses, and social interactions reward consistency over months. Three reward horizons stacked on the same single daily action, so there is always a reason to open the app today and a reason to come back tomorrow.

"One mood. One mission. One walk. Every day — and your dog's avatar (and your dog) both get a little better for it."